

Long Term Water Sustainability

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Erdős Institute Data Science Bootcamp
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Climate Change and Water Availability

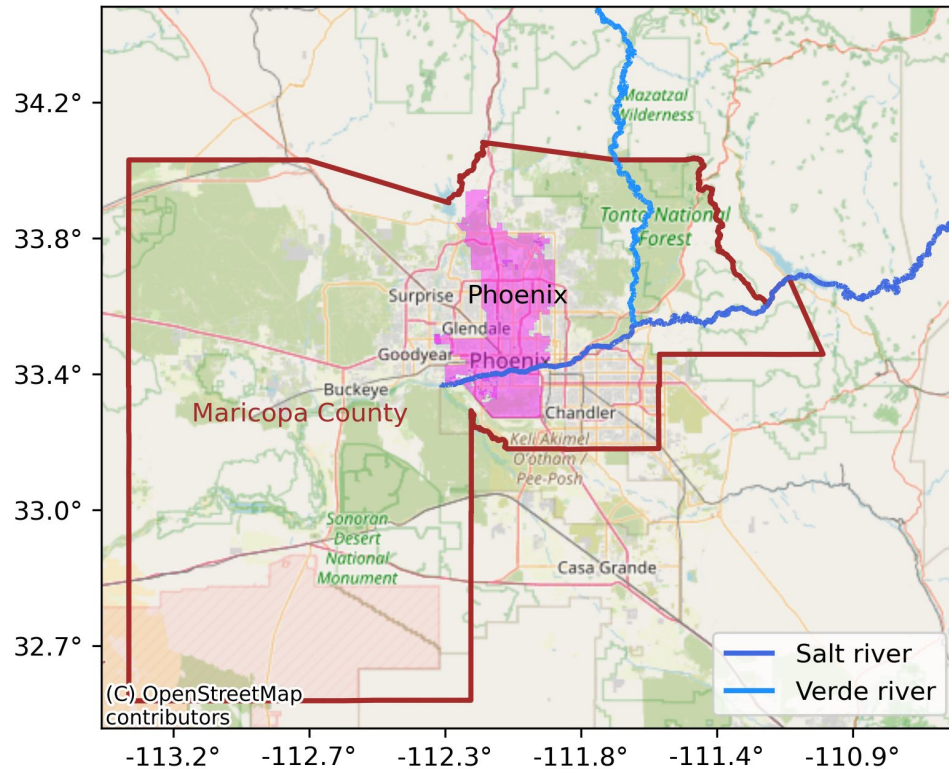
Climate change threatens water availability in many places worldwide.

The Southwest United States in particular faces imminent water shortage issues.

In this project, we investigated sustainability and future availability of water consumption in Maricopa County, Arizona.

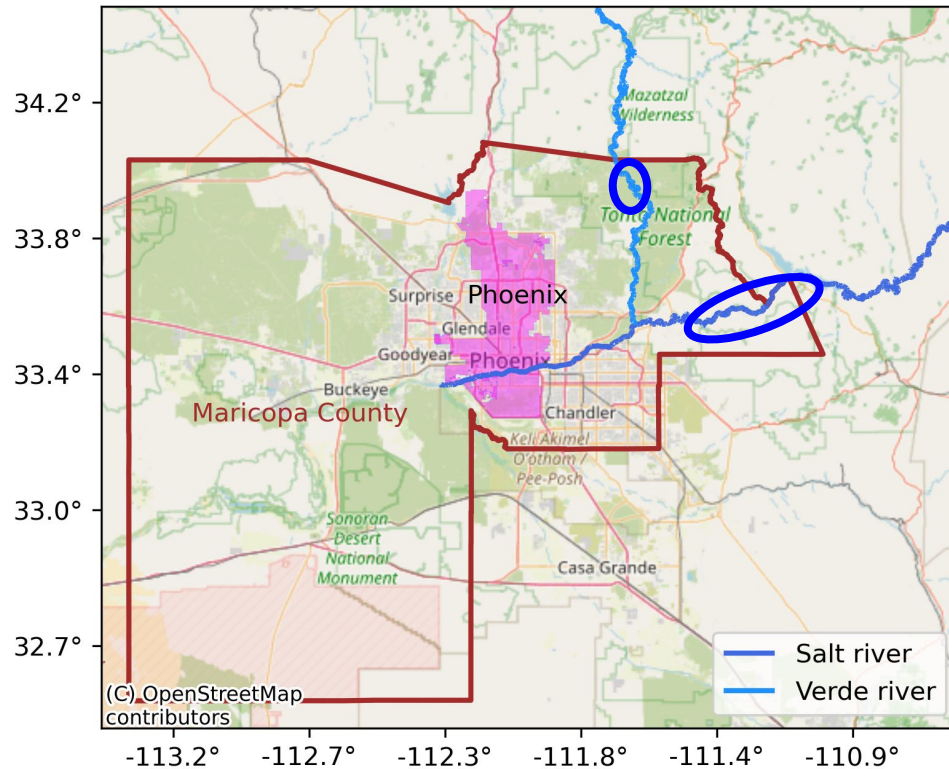
Maricopa County

Population: 4.4 million



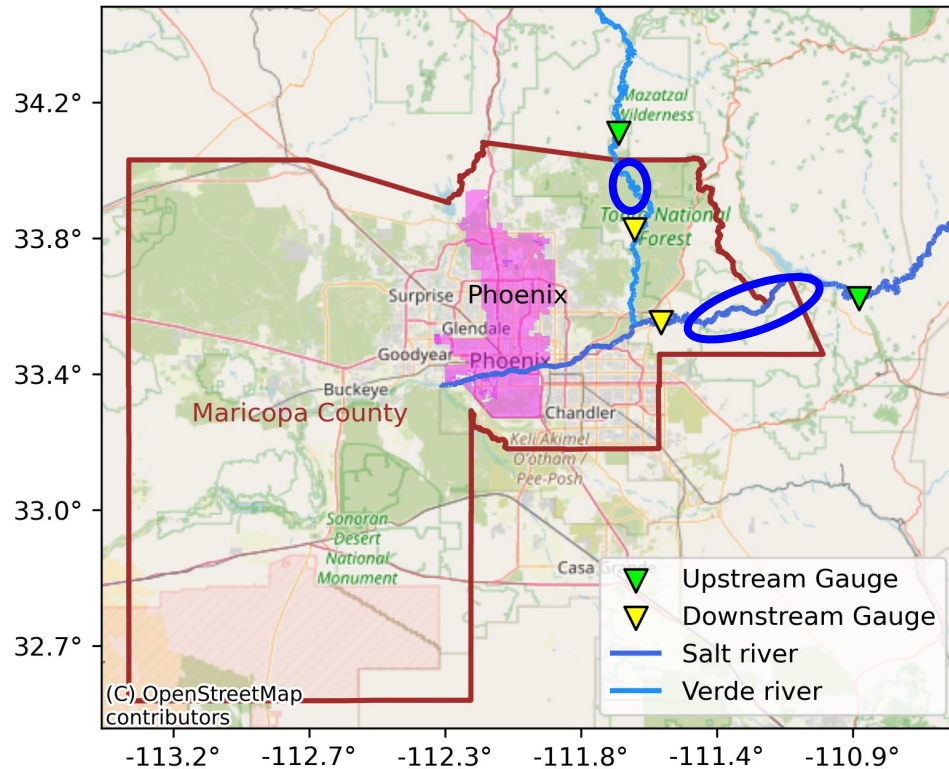
Maricopa County

Salt River Project Reservoirs



Maricopa County

Salt River Project Reservoirs



Variables and Data Sources

Response River flow rates: USGS Water Data



Features

Climate

Historical: NOAA
nClimGrid



Future: LOCA2
CMIP6 projections,
Climate change
scenarios - **SSP2-4.5**
(moderate) & **SSP5-8.5**
(high)



Human population

Historical: US Census
Bureau, 
National Bureau of
Economic Research, 
National Historical GIS 

Future: Arizona Office of
Economic Opportunity



Irrigated farmland area

USDA census of
agriculture



Data Centers

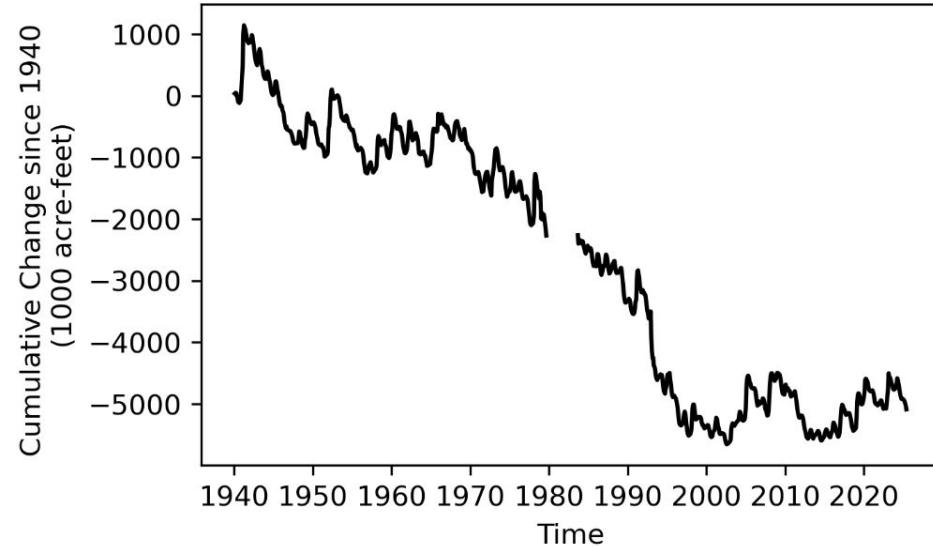
(number and total
capacity):
Cleanview.co



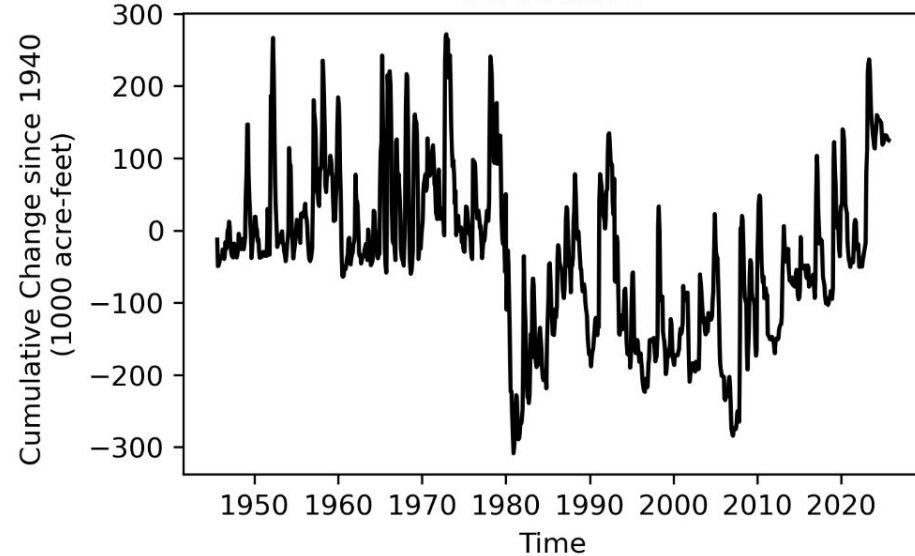
Change in reservoir water storage

Cumulative sum of upstream rate minus downstream rate

Salt River



Verde River



Verde River Flow Modeling

Features

Upstream Flow:

Climate:

- Watershed temp and precipitation

River:

- Selected upstream lags

Downstream Flow:

Climate:

- Watershed temp and precipitation
- Maricopa county temp and precipitation

Human Demand:

- Population (Maricopa County)
- Irrigated Farmland (Maricopa County)
- Operational data centers number and MW (Maricopa County)

River:

- Upstream flow
- Selected downstream lags

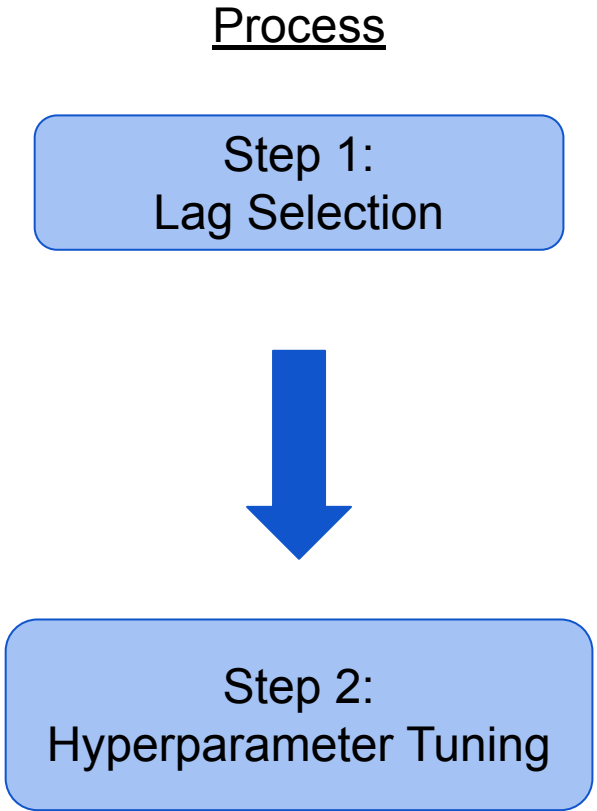
Model Selection

Training set: historical data 1945-2017

- Models:**
- Random forest
 - AdaBoost
 - Gradient boost
 - XGBoost

Evaluation Method: 5-fold time series cross validation

KPI: Average fold RMSE (minimize)



Upstream Best Model

Gradient Boost

- Lags 1 and 11 months
- max_depth = 3
- n_estimators = 100

Evaluation on Test Set

Test set: historical data 2018-2025

Model	RMSE	% of Mean Log Flow
Gradient Boost (selected model)	0.33	6.1%
Seasonal Average (baseline)	0.75	13.9%

Gradient boost reduces forecast RMSE by 56% relative to the baseline.

Downstream Best Model

XGBoost

- Lags 5 and 6 months
- max_depth = 6
- n_estimators = 100

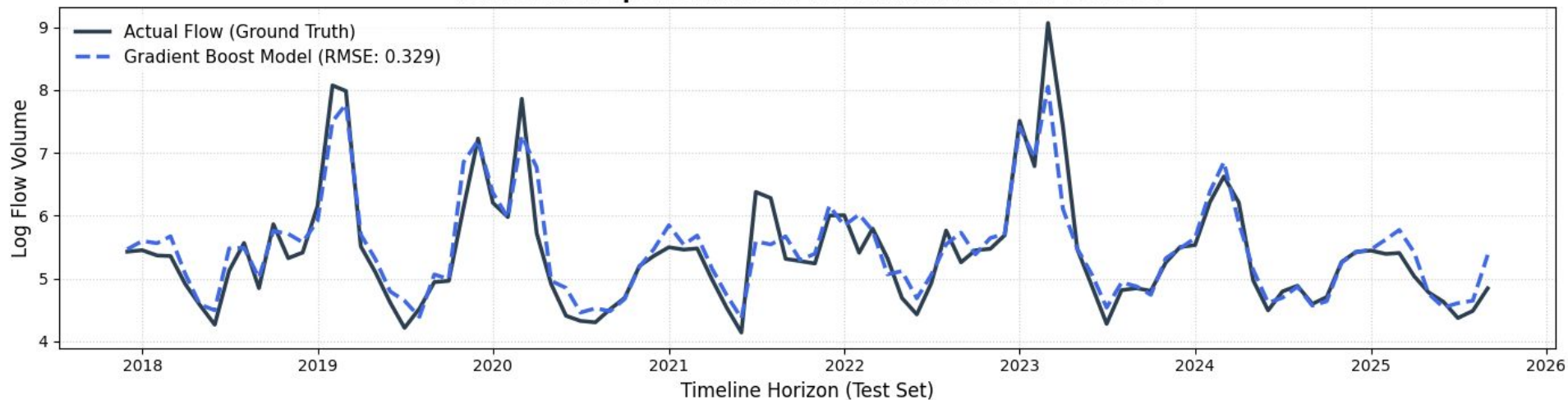
Evaluation on Test Set

Test set: historical data 2018-2025

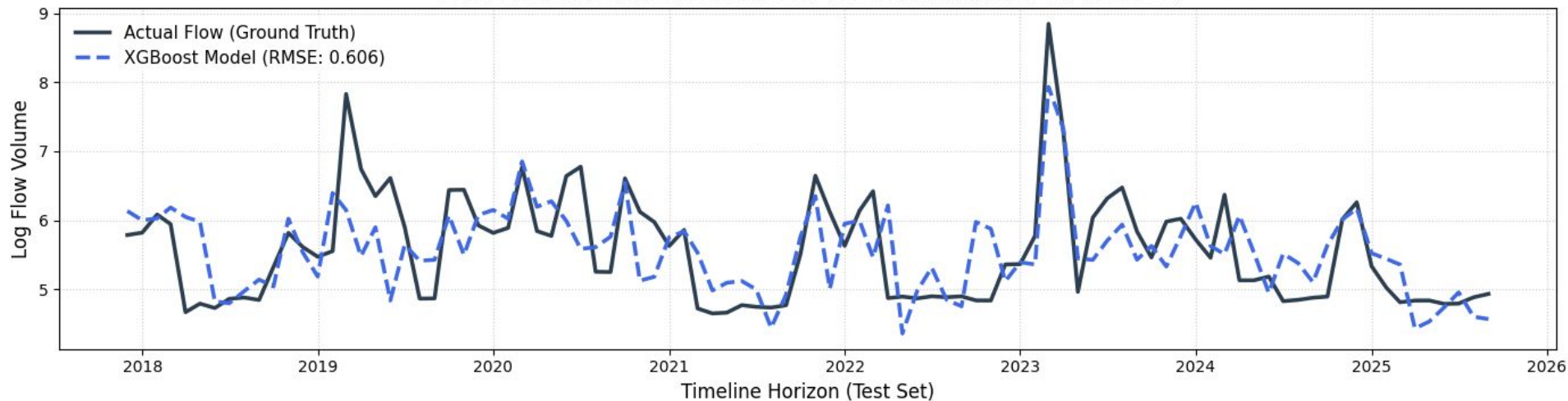
Model	RMSE	% of Mean Log Flow
XGBoost (selected model)	0.61	10.9%
Seasonal Average (baseline)	0.75	13.5%

XGBoost reduces forecast RMSE by 19% relative to the baseline.

Verde River Upstream Flow: Model Evaluated on Test Set



Verde River Downstream Flow: Model Evaluated on Test Set



Salt River Flow Modeling

Features — Salt River

Downstream Data

Climate

- Watershed temp & precipitation
- Maricopa County temp & precipitation
- Population (Maricopa County)
- Irrigated Farmland (Maricopa County)
- Operational Data Centers — Number & MW
- Upstream Flow (same month)
- Selected lags: 9 and 12 months

Upstream Data

Climate

- Watershed temp & precipitation
- Selected lags: 1 and 12 months

Model Selection — Salt River

Process

Training set:

1914-2017 (upstream) · 1935-2017
(downstream)

Step 1: Lag Selection

Greedy search over lags 1-12 months

Step 2: Hyperparameter Tuning

Grid search over tree depth & count

Models Compared

- Random Forest
- XGBoost

Evaluation Method

3-fold recursive time-series
cross-validation

KPI

RMSE (minimize)

Upstream Data Results — Salt River

Evaluation on Test Set

Test set: historical data 2017-2025

Winning Model:

Random Forest

Lags 1 and 12 months

max_depth = 10

n_estimators = 300

Model	RMSE	% of Mean Flow
Random Forest (selected)	0.55	8.93%
Seasonal Average (baseline)	1.08	17.65%

49.4%

reduction in forecast error vs. the baseline

Downstream Data Results — Salt River

Winning Model:

Random Forest

Lags 9 and 12 months

max_depth = 6

n_estimators = 100

Evaluation on Test Set (One Month Ahead)

Test set: historical data 2017-2025

Random Forest RMSE: 1.20

21.02% of mean log flow

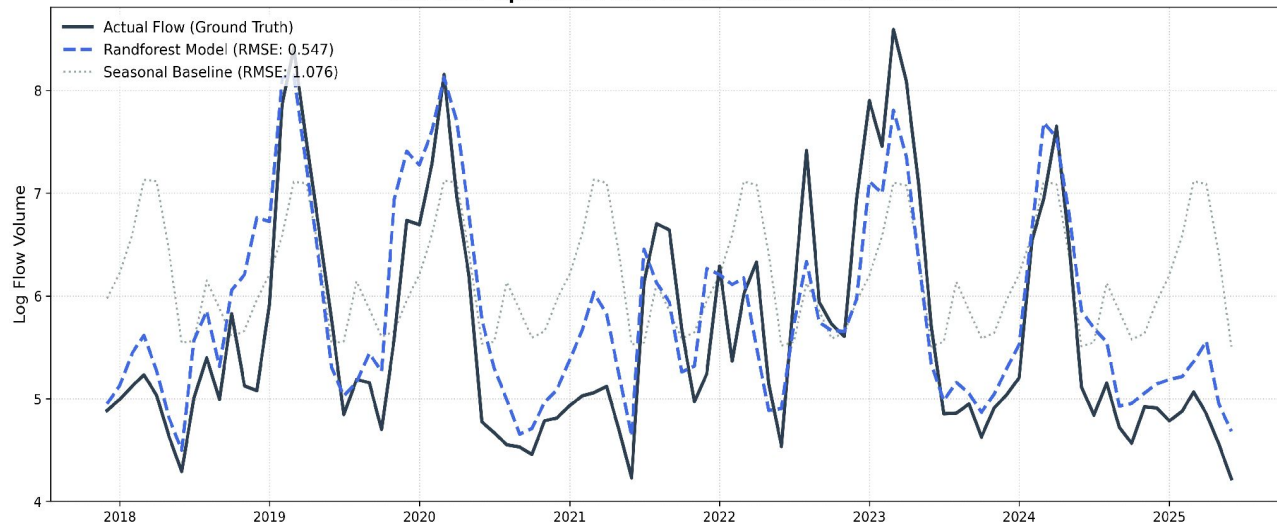
Seasonal Average RMSE: 1.23

21.63% of mean log flow

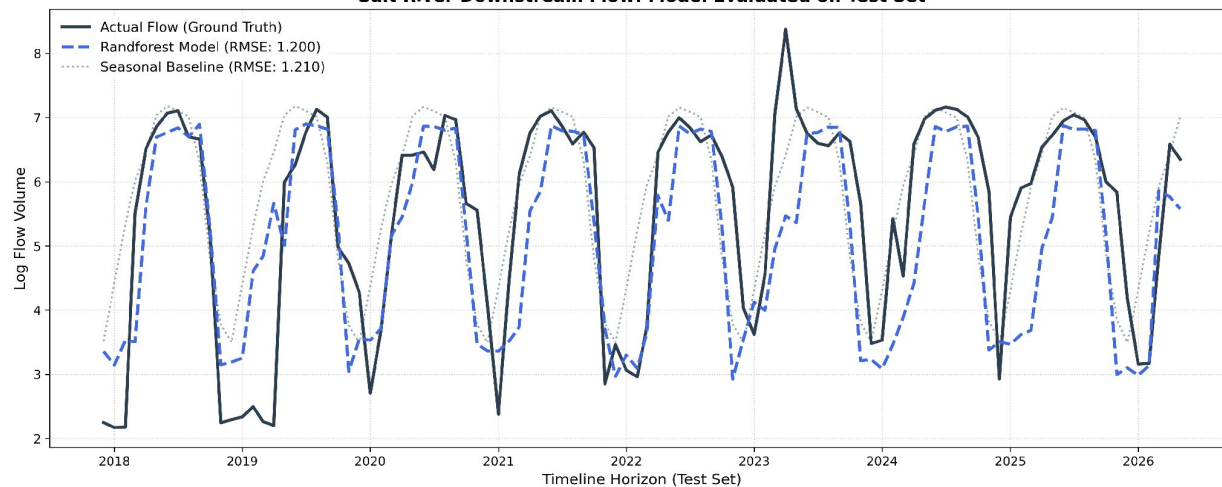
2.8%

reduction in forecast error vs. the baseline

Salt River Upstream Flow: Model Evaluated on Test Set



Salt River Downstream Flow: Model Evaluated on Test Set



Future Predictions for Water Storage Changes

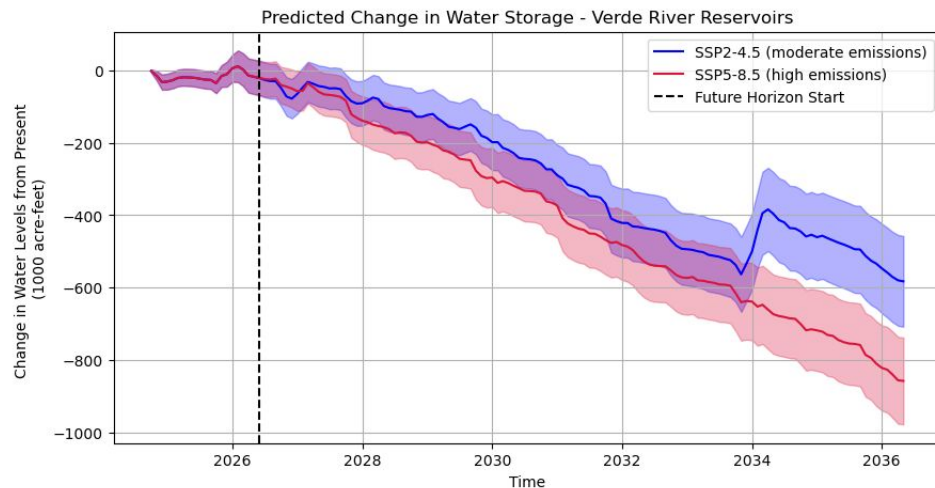
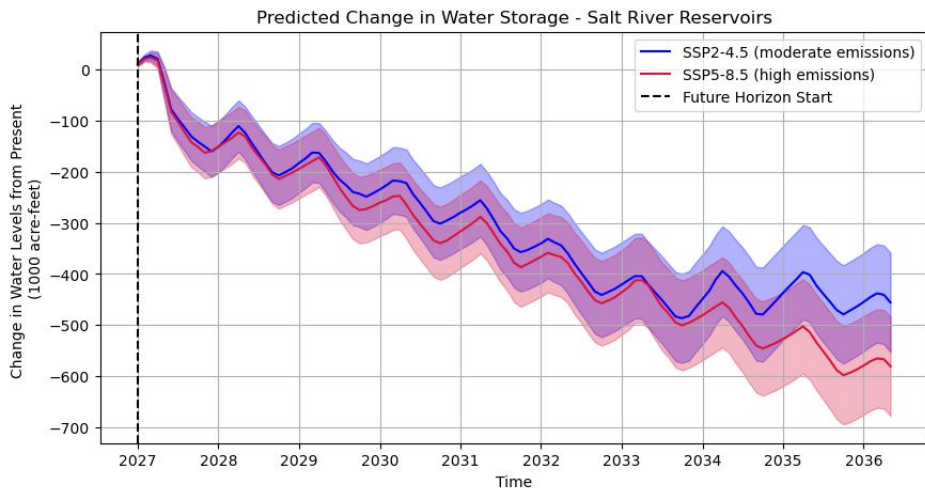
What are our predictions for the changes in water level storage in the Salt River Project reservoirs?

Future climate data predictions: Shared Socio-economic Pathways (SSP)

Two scenarios:

- Moderate emissions: SSP2-4.5, a “middle of the road” projection for future development
- High emissions: SSP5-8.5, a pessimistic projection with continued fossil-fuel development

Future Predictions for Water Storage Changes



Total capacity of the Salt River Project reservoirs is about 3.2 million acre-feet

Predicted rate of water depletion far exceeds rates seen from 2000-2020

The predicted decrease in the next 10 years is a significant fraction of the total reservoir capacity

Limitations

- Reservoir water levels were estimated from river flow rates
- Relationship between features and response may change over time - our models assumed fixed relationship
- Did not account for uncertainty in future values of features
- Compounding uncertainties in future predictions grow farther into the future

Conclusions and Policy Recommendations

Conclusions

- Water levels in the Salt and Verde river reservoirs are expected to decline in the near future.

Policy recommendations

- Raising public awareness
- Stronger regulation of water use
- Further research and monitoring



Thank you!